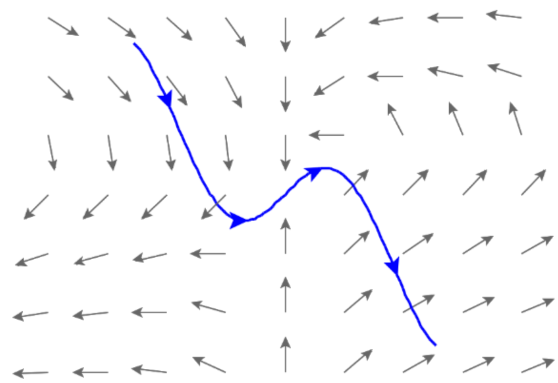
MTH 202 Chapter 5: Topics In Vector Calculus

5.2.2 Work Integrals



Motivation: Suppose, as in the last section, we have a curve C that we are interested in travelling along, but we are now wanting to move along this curve through a vector field.

In the last section we saw that a line integral was useful for "adding up" a quantity as we progressed along a path. In this section we are interested in the vector field's effect on the motion of an object along the curve C, specifically in regards to the amount of work done by the vector field on any object that traverses C.

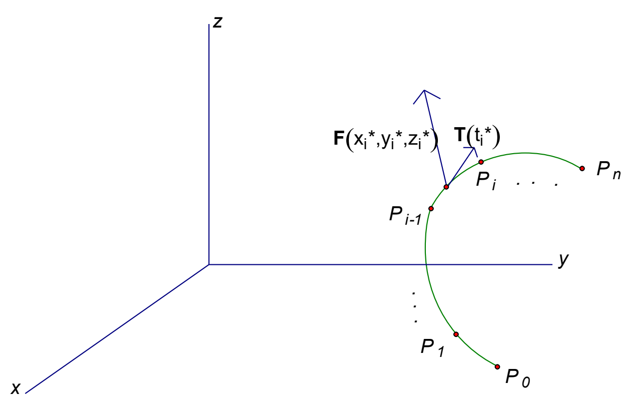
When the vector field is moving with the direction of the curve it contributes positive work in the motion of an object along the curve. When the vector field is moving against the direction of the curve it contributes negative work in the motion of an object along the curve.

Exploration: Suppose that $\mathbf{F}(x,y,z)=\langle P(x,y,z),Q(x,y,z),R(x,y,z)\rangle$ is a continuous vector field. Suppose also that $\mathbf{F}(x,y,z)$ models some force field in space (for example $\mathbf{F}$ could represent a gravitational field or an electric force field).

Suppose a particle moves through this force field along a smooth curve C which is given by the parameterization $\mathbf{r}(t)$ where $a \le t \le b$. Suppose that the particle starts its movement at the point P_0 and ends its movement at the point P_n. We want to know the amount of work done by $\mathbf{F}$ on the particle as it moves from P_0 to P_n.

This problem is particularly complicated because the force is continuously changing as the particle moves through space. So, we start by trying to approximate the force in hopes of being able to relate our approximation to an integral.

To do this we partition $[a,b]$ into n equally spaced intervals. This inherently partitions the curve C with $n+1$ points. A picture is below:



We let Δs_i represent the length of the segment from P_{i-1} to P_i (the segment approximates C from P_{i-1} to P_i).

We let t_i^* be some sample point in $[t_{i-1},t_i]$, and suppose that $\mathbf{r}(t_i^*)=\langle x_i^*,y_i^*,z_i^*\rangle$.

We approximate the work done by $\mathbf{F}$ on the particle as the particle moves from P_{i-1} to P_i as follows. As long as P_{i-1} and P_i are close together, the force is approximately $\mathbf{F}(x_i^*,y_i^*,z_i^*)$ throughout the movement from P_{i-1} to P_i.

Moreover, as long as P_{i-1} and P_i are close together, the motion of the particle is approximately in the direction of the unit tangent vector **T** to the curve at any point between P_{i-1} and P_i, and the distance the particle travels from P_{i-1} to P_i is approximately Δs_i.

So, by the way work is defined, the work done by **F** on the particle as the particle moves from P_{i-1} to P_i is approximately:

$$\mathbf{F}\left(x_i^*, y_i^*, z_i^*\right)\cdot \mathbf{T}\left(t_i^*\right)\Delta s_i.$$

Thus, the total work done by the force **F** in moving the particle along *C* is:

$$W = \lim_{n\to\infty}\sum_{i=1}^{n}\mathbf{F}\left(x_i^*, y_i^*, z_i^*\right)\cdot \mathbf{T}\left(t_i^*\right)\Delta s_i = \int_C \mathbf{F}\cdot\mathbf{T}\,ds$$

We know that $ds = \sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2}\,dt = \left\|\mathbf{r}'(t)\right\|dt$, and from chapter 2 we know that $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\left\|\mathbf{r}'(t)\right\|}$.

So, we have:

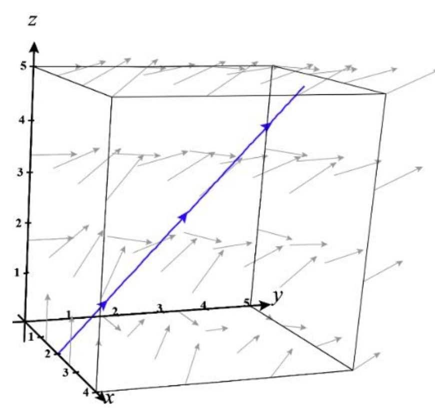
$$\int_C \mathbf{F}\cdot\mathbf{T}\,ds = \int_a^b \left(\mathbf{F}(\mathbf{r}(t))\cdot\frac{\mathbf{r}'(t)}{\left\|\mathbf{r}'(t)\right\|}\right)\left\|\mathbf{r}'(t)\right\|dt = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt.$$

Definition: Let **F** be a continuous vector field defined on a smooth curve *C* given by a vector function $\mathbf{r}(t)$ where $a \le t \le b$. Then, the <u>line integral of **F** along *C*</u> is

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \mathbf{F}\cdot\mathbf{T}\,ds = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt$$

*Note: We will address an intuitive meaning of d***r** *in the next example. One should note that* **F** *may have any number of components and the above definition will still apply.*

<u>**Example 1:**</u> Find the work done by the force field $\mathbf{F} = \langle y, z, x\rangle$ in moving a particle along the line segment, *C*, from $(2,0,0)$ to $(3,4,5)$. (Notice this is the same as Ex 5 in section 5.2.1!)



Using techniques from Chapter 2 we note that a parameterization for *C* is $\mathbf{r}(t) = \langle 2+t, 4t, 5t\rangle$ where $0 \le t \le 1$. We know that the work is equal to the line integral of **F** along *C*.

So, by the above definition we have that

$$W = \int_0^1 \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt = \int_0^1 \mathbf{F}((2+t,4t,5t))\cdot\langle 1,4,5\rangle\,dt = \int_0^1 \langle 4t, 5t, 2+t\rangle\cdot\langle 1,4,5\rangle\,dt$$

$$= \int_0^1 (4t+20t+10+5t)\,dt = \frac{49}{2}.$$

Observation:

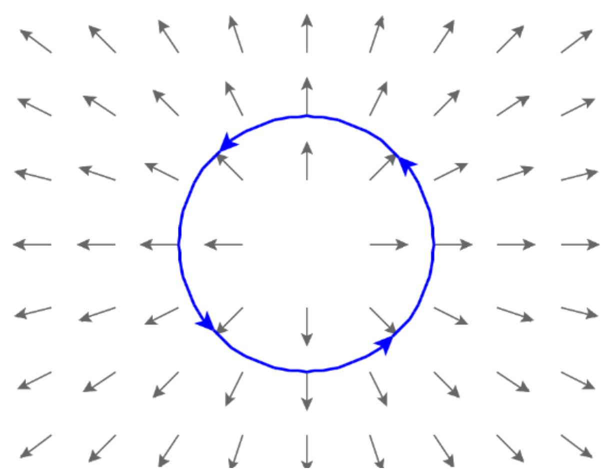
We can view our resulting work integral as $W = \int_C y\ dx + z\ dy + x\ dz$.

If we think of $d\mathbf{r} = \langle dx, dy, dz \rangle$, we see that $W = \int_C y\ dx + z\ dy + x\ dz = \int_C \langle y, z, x \rangle \cdot \langle dx, dy, dz \rangle = \int_C \mathbf{F} \cdot d\mathbf{r}$.

So, thinking of $d\mathbf{r}$ as $\langle dx, dy, dz \rangle$ gives us a way to interpret the first part of the above definition, and we now see that line integrals with respect to *x, y,* and *z* do indeed show up when we need to find work!

Note: One may recognize $\int_C y\ dx + z\ dy + x\ dz$ *as the sum of line integrals we found for the last example in the previous section.*

Example 2: Find the work done by $\mathbf{F}(x, y) = \langle x, y \rangle$ on a particle which moves along the curve C given by $\mathbf{r}(t) = \langle \cos(t), \sin(t) \rangle$ where $0 \le t \le \pi$.



We know that the work is

$$W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \langle x, y \rangle \cdot \langle dx, dy \rangle = \int_C xdx + ydy = \int_C xdx + \int_C ydy.$$

We note that parametric equations for the curve are

$x = \cos(t) \quad y = \sin(t)$.

These imply that $dx = -\sin(t)dt$ and $dy = \cos(t)dt$.

Thus, we have that

$$\int_C xdx + \int_C ydy = -\int_0^\pi \cos(t)\sin(t)dt + \int_0^\pi \cos(t)\sin(t) = 0.$$

Note: One should draw a picture of the particle's path and this vector field to see why the work is equal to 0. When this is done you can see that the at all points the vector field is orthogonal to the curve C and therefore $\mathbf{F} \cdot \mathbf{T} = 0$ *at all points*. This means that $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (\mathbf{F} \cdot \mathbf{T})ds = \int_C (0)ds = 0$.

Note: One could have solved the above problem with $W = \int_0^\pi \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)dt$ *as well and gotten a result as follows.*

$$\begin{aligned} W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \langle x, y \rangle \cdot \langle dx, dy \rangle \\ &= \int_0^\pi \langle \cos(t), \sin(t) \rangle \cdot \langle -\sin(t), \cos(t) \rangle dt \\ &= \int_0^\pi \left[-\cos(t)\sin(t) + \cos(t)\sin(t) \right] dt = \int_0^\pi 0\, dt = 0 \end{aligned}$$